Let's look at several of these basis functions in detail. Figure (a) shows the cosine wave $c_0[\,]$. This is a cosine wave of zero frequency, which is a constant value of one. This means that $ReX[0]$ holds the average value of all the points in the time domain signal. In electronics, it would be said that $ReX[0]$ holds the **DC offset**. The sine wave of zero frequency, $s_0[\,]$, is shown in (b), a signal composed of all *zeros*. Since this can not affect the time domain signal being synthesized, the value of $ImX[0]$ is *irrelevant*, and always set to zero. More about this shortly.

Figures (c) & (d) show $c_2[\,]$ & $s_2[\,]$, the sinusoids that complete *two* cycles in the N points. These correspond to $ReX[2]$ & $ImX[2]$, respectively. Likewise, (e) & (f) show $c_{10}[\,]$ & $s_{10}[\,]$, the sinusoids that complete *ten* cycles in the N points. These sinusoids correspond to the amplitudes held in the arrays $ReX[10]$ & $ImX[10]$. The problem is, the samples in (e) and (f) no longer *look* like sine and cosine waves. If the continuous curves were not present in these graphs, you would have a difficult time even detecting the pattern of the waveforms. This may make you a little uneasy, but don't worry about it. From a mathematical point of view, these samples do form discrete sinusoids, even if your eye cannot follow the pattern.

The highest frequencies in the basis functions are shown in (g) and (h). These are $c_{N/2}[\,]$ & $s_{N/2}[\,]$, or in this example, $c_{16}[\,]$ & $s_{16}[\,]$. The discrete cosine wave alternates in value between 1 and -1, which can be interpreted as sampling a continuous sinusoid at the *peaks*. In contrast, the discrete sine wave contains all zeros, resulting from sampling at the *zero crossings*. This makes the value of $ImX[N/2]$ the same as $ImX[0]$, always equal to zero, and not affecting the synthesis of the time domain signal.

Here's a puzzle: If there are N samples entering the DFT, and $N+2$ samples exiting, where did the extra information come from? The answer: two of the output samples contain *no* information, allowing the other N samples to be fully independent. As you might have guessed, the points that carry no information are $ImX[0]$ and $ImX[N/2]$, the samples that always have a value of zero.

Synthesis, Calculating the Inverse DFT

Pulling together everything said so far, we can write the **synthesis equation**:

$$x[i] = \sum_{k=0}^{N/2} Re\bar{X}[k]\cos(2\pi ki/N) + \sum_{k=0}^{N/2} Im\bar{X}[k]\sin(2\pi ki/N)$$

EQUATION 8-2
The synthesis equation. In this relation, $x[i]$ is the signal being synthesized, with the index, i, running from 0 to $N-1$. $Re\bar{X}[k]$ and $Im\bar{X}[k]$ hold the amplitudes of the cosine and sine waves, respectively, with k running from 0 to $N/2$. Equation 8-3 provides the normalization to change this equation into the inverse DFT.